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MANUFACTURING & SERVICE OPERATIONS MANAGEMENT

Submitted to *Manufacturing & Service Operations Management*

Vol. 00, No. 0, Xxxxx 0000, pp. 000–000

ISSN 1523-4614, EISSN 1526-5498

When Front-Stage Innovation Overloads Back-Stage Workers: A Dynamic Model of Workforce, Capability, and Customer Trust in Omnichannel Retail

(Authors' names are not included for peer review)

Abstract. Problem definition. Digital investment in omnichannel retail generates back-stage demand on workers that standard dashboards cannot see. Front-stage automation amplifies operational complexity faster than back-stage capacity can absorb it, creating a structural measurement gap: technology returns and labour consequences appear on different ledgers, and the workforce that determines whether technology delivers its designed capability goes systematically underinvested.

Methodology/results. We formalise this gap as a parsimonious dynamic model and derive three analytically characterised results. (i) A computable digitalisation ceiling exists beyond which additional front-stage investment destroys customer lifetime value by overloading back-stage workers; revenue metrics continue rising even as lifetime value collapses. (ii) The entire return to workforce investment flows through service capacity, invisible to dashboards that treat payroll as a cost line. (iii) The system is bistable: at our baseline calibration, the neglected firm faces a safe rollout window approximately 37% the size of its investing peer. Three empirical strategies provide directionally consistent evidence across different identification strategies. A statistically significant inverted-U between e-commerce share and customer satisfaction, and a post-shock decline in the American Customer Satisfaction Index following major omnichannel launches, support the digitalisation ceiling. County-level wage and employment data from the Bureau of Labor Statistics ($N = 30{,}951$ county–year observations across 3,115 US counties, 2015–2024) support a labour-market analogue of the Good Jobs Buffer: counties with higher initial retail wages show significantly more resilient employment as e-commerce grows ($p = 0.025$).

Managerial implications. The model supplies two computable numbers absent from standard dashboards: a digitalisation ceiling and a maximum safe rollout size. Both are workforce-contingent, making workforce investment a structural requirement of digital strategy rather than a social concession.

Funding: No external funding was received for this research.

Key words: omnichannel retail, workforce well-being, service capability, customer trust, system dynamics, people-centric operations

1. Introduction

Omnichannel grocery retailers have invested heavily in digital capabilities in recent years (Hübner et al. 2021, Bijmolt et al. 2021), enabling customers to build baskets across channels, reserve items for same-day collection, and track substitutions in real time. Yet as digital engagement has grown, retailers have encountered a structural asymmetry in how costs and returns are measured. Technology investment registers as successful in the front-stage domain where it is evaluated, while the back-stage workforce that must fulfil the digitally generated demand absorbs consequences that appear nowhere in the technology business case (Corbett 2024).

The asymmetry is structural, not incidental. Front-stage digital capabilities generate customer journeys that multiply handoff complexity far faster than order volume grows (Saghiri et al. 2017), placing distinctive operational demands on back-stage workers (pickers, drivers, click-and-collect associates). The empirical literature confirms the scale of these consequences: workforce investment generates returns that are both large and invisible to standard metrics (Fisher et al. 2021, Kesavan et al. 2022), and firms that invest in workforce capacity outperform on both operational and financial measures (Ton 2014), yet the causal chain runs through mechanisms that standard dashboards do not monitor.

What is missing is a formal account of the mechanism. The empirical literature establishes that workforce investment pays off; it does not characterise *when* additional front-stage investment begins to destroy rather than create value, *how fast* a firm can scale digitalisation before the back-stage workforce is overwhelmed, or *what dynamic relationship* connects technology investment, workforce well-being, and customer trust over time. These are not questions that regression analysis on historical data can answer: they require a model that treats workforce well-being and customer trust as co-evolving state variables responding to a controllable investment sequence (see §2).

This paper formalises the economic structure of this problem as a parsimonious dynamic model. Technology and workforce are not independent inputs to service capability but strategic complements (Milgrom and Roberts 1990, Bainbridge 1983, Kremer 1993). The central modelling innovation is the *workforce capacity multiplier*: effective service capability is the product of the technology stock and a workforce engagement index. When front-stage demand grows faster than back-stage workforce capacity, a service gap opens that depletes workforce well-being, which reduces effective capability, which widens the gap further: a reinforcing deterioration that single-domain models cannot generate.

Three formally characterised results follow. A *Front-Stage Trap* (Theorem 1): a computable digitalisation threshold exists beyond which additional front-stage investment reduces customer lifetime value by overloading back-stage workers. A *Good Jobs Buffer* (Theorem 2): the return to workforce investment flows through a capacity-level channel that is invisible to dashboards measuring only operational sensing. A *Burnout Cliff* (Theorem 3): the system exhibits bistability, and at baseline calibration, a firm that neglects its workforce faces a safe deployment window approximately 37% the size of one that invests in its people. A fourth result, Proposition 4, establishes that the marginal value of workforce investment is increasing in digitalisation level. Workforce investment is therefore not a social concession but a structural requirement of digital strategy.

2. Related Literature

This paper sits at the intersection of three research streams. We review each and identify the gaps the model addresses.

Service quality erosion. The foundational insight that service quality can erode endogenously under demand pressure originates with Oliva and Sterman (2001), whose system-dynamics model (Sterman 2000) produces death spirals in which workers cut corners under demand pressure. Their finding that workers are roughly twice as willing to cut corners as to work overtime grounds the overload-asymmetry parameter σ in the present model. Subsequent work extends the logic to capability erosion (Rahmandad and Repenning 2016, Repenning and Sterman 2001), e-commerce growth (Oliva et al. 2003), and regulatory compliance (Martinez-Moyano et al. 2014); KC and Terwiesch (2009) confirm that sustained high workload degrades service outcomes empirically, and Corbett (2024) calls for formal dynamic models of workforce well-being in operations; KC and Terwiesch (2009) provide empirical evidence that sustained workload degrades outcomes, underscoring the need for such models. **Gap 1:** no model in this tradition treats workforce well-being as a positive capability multiplier. **Gap 2:** no Oliva–Sterman extension adds customer trust, a front-/back-stage split, or digital scaling.

Workforce investment and retail performance. A substantial empirical literature documents the workforce-performance link. The Good Jobs Strategy (Ton 2014) identifies four operational choices that drive outperformance; causal evidence shows that responsible scheduling raises productivity (Kesavan et al. 2022) and that profit-maximising staffing exceeds prevailing practice (Fisher et al. 2021, Dubé et al. 2020). The conceptual mechanism is the *service profit chain* (Heskett et al. 1994), which has not been formalised as a dynamic model with computable boundary conditions. The Job Demands-Resources model (Bakker and Demerouti 2007, Demerouti et al. 2001) provides the micro-foundation: overload depletes well-being faster than equivalent surplus restores it, and productivity responses scale with engagement (Mas and Moretti 2009). These two features (asymmetric overload and engagement-scaled sensitivity) underpin the workforce dynamics in §3.3. **Gap 3:** no dynamic model formalises the Good Jobs Strategy or the service profit chain with computable boundary conditions.

Omnichannel capability. Digitalisation creates considerable challenges for retail logistics, with fulfilment complexity scaling super-linearly as channel combinations increase (Hübner et al. 2021, Saghiri et al. 2017, Bell et al. 2018, Acimovic and Graves 2015). Automation shifts the operator's role toward exception handling (Bainbridge 1983), making the quality of residual human tasks more consequential for overall output (Gans and Goldfarb 2026). Technology–workforce strategic complementarity (Milgrom and Roberts 1990, Aral et al. 2012) motivates the model's central structural assumption that the marginal value of workforce engagement rises with the technology base (§3.1). **Gap 4:** no model formalises front-to-back cross-domain overload. **Gap 5:** the digitalisation-workforce cross-partial has not been derived for services.

Table 1 summarises the five gaps and the model's response to each.

3. Model

Sections 3.1–3.5 formalise the workforce capacity multiplier introduced in §1. The front-stage digitalisation level $F > 0$ is treated as an exogenous managerial control: it sets the complexity and volume of customer-facing digital features, and thereby determines the demand placed on the back-stage operation. The model traces the dynamic consequences of choosing F for the three co-evolving back-stage state variables and, through them, for customer lifetime value (CLV). Each of the eight free parameters has a single, distinct

Table 1 Literature gaps and model contributions.

Gap	Closest prior work	This paper
1. No model treats W as a positive capability multiplier	Oliva and Sterman (2001): W as degradation only	Effective capability as the product of technology and workforce engagement
2. No Oliva–Sterman extension adds trust, domain split, or digital scaling	Oliva and Sterman (2001); Martinez-Moyano et al. (2014)	Customer trust as a co-evolving state; front/back domain split; digital scaling
3. Good Jobs Strategy / service profit chain (SPC) unformalised	Ton (2014); Heskett et al. (1994): empirical only	Dynamic formalisation with analytical equilibria and boundary conditions
4. No cross-domain front-to-back overload model	Saghiri et al. (2017): conceptual only	Super-linear back-stage demand as a function of digital depth and customer trust
5. Digitalisation-workforce cross-partial not derived for services	No direct service prior; see also Gans and Goldfarb (2026) (O-ring)	Formally derived: the return to workforce investment rises with the technology base

role with a clean kill condition (Table 2): setting it to its neutral value eliminates exactly one mechanism, allowing the model to nest single-domain benchmarks as special cases.

3.1. Stocks and Effective Capability

Three state variables evolve jointly: $K(t) > 0$, the back-stage technology stock (installed infrastructure, warehouse management systems, and process standardisation); $W(t) \in (0, 1)$, an index of workforce well-being (engagement, energy, and cognitive capacity among pickers, drivers, and click-and-collect associates); and $T(t) \in (0, 1)$, customer trust (the customer base's aggregate confidence that the retailer will deliver reliably). All three are dimensionless and normalised to their natural operating ranges. Throughout, starred variables (K^*, W^*, T^*) denote steady-state equilibrium values; F_0 is the pre-shock digitalisation level; $\Delta F = F - F_0$ is the upgrade step.

The central structural innovation is the *effective capability*. A warehouse management system or substitution-handling process can only deliver its designed performance if the workforce operating it has the capacity to do so. Effective capability is therefore not the technology stock alone but its product with a workforce engagement multiplier:

$$K_{\text{eff}}(t) \,=\, K(t)\cdot h\big(W(t)\big),$$
$$h(W) = (1-p) + pW, \quad p \in (0,1). \tag{1}$$

Effective capability is the product of the technology stock and a workforce capacity multiplier $h(W)$, ranging from $h(0) = 1 - p$ (zero well-being: only the automation-driven fraction is realised) to $h(1) = 1$ (full design capacity). The cross-partial

$$\frac{\partial^2 K_{\text{eff}}}{\partial K\,\partial W} \,=\, p \,>\, 0 \tag{2}$$

is strictly positive for all $p > 0$, establishing technology and workforce as complements in the Milgrom and Roberts (1990) sense.

The parameter p governs complementarity: $p = 0$ decouples workforce from capability; $p = 1$ makes the operation fully labour-intensive. The baseline $p = 0.5$, motivated by capital-skill complementarity evidence (Krusell et al. 2000), represents a moderate starting point; estimation is discussed in §7.2. All three theorems hold qualitatively for any $p \in (0,1)$.

3.2. Demand and the Service Gap

Customer demand on back-stage operations is

$$D(T,F) \,=\, (1 + aT)\cdot F^{b}, \tag{3}$$

where $a > 0$ is the trust-demand coupling and $b > 1$ is the front-stage amplification exponent. The trust-demand coupling reflects the established finding that customer trust drives spending consolidation and order frequency (Sirdeshmukh et al. 2002, McKnight et al. 2002). The super-linearity $b > 1$ captures the well-documented increase in back-stage fulfilment complexity as front-stage channel breadth grows: as shown in §2, Bell et al. (2018) and Acimovic and Graves (2015) document patterns consistent with fulfilment error rates and demand-capacity mismatches rising faster than linearly with channel combinations. We set $b = 1.3$ as an illustrative, structurally conservative baseline motivated by this literature; the Front-Stage Trap requires only $b > 1$, and all three theorems hold qualitatively across $b \in [1.1, 2.0]$ (EC Table EC.3).

The *service gap* measures the shortfall between demand and effective capability:

$$G(t) \,=\, D\big(T(t),\, F\big) \,-\, K(t)\cdot h\big(W(t)\big). \tag{4}$$

When $G > 0$ (overload), customers experience failures. When $G < 0$ (surplus), service runs reliably. We write $[G]^+ = \max(G, 0)$ and $[-G]^+ = \max(-G, 0)$ for the positive and negative parts.

3.3. Dynamics

The feedback loops in §3.5 are built from the three ordinary differential equations (ODEs) introduced below, which translate each economic mechanism into a rate equation. All rate parameters carry units of months^{-1}; each ODE is piecewise-smooth, activating different terms in the overload regime ($G > 0$, so $[G]^+ = G$, $[-G]^+ = 0$) and the surplus regime ($G < 0$, so $[G]^+ = 0$, $[-G]^+ = -G$).

Capability. The technology stock evolves as

$$\frac{dK}{dt} \,=\, r\,[G]^+\,W \,-\, \delta\,K \,+\, u, \tag{5}$$

where $r > 0$ is the sensing-adaptation rate, $\delta > 0$ is the depreciation rate, and $u \geq 0$ is direct back-stage investment. The sensing term $r\,[G]^+\,W$ captures organisational learning from service failures (Tucker 2007): the product structure requires both a gap to detect ($[G]^+ > 0$) *and* workforce capacity to act on it ($W > 0$), reflecting the finding that capability accumulation scales with the workforce knowledge stock (Argote and Epple 1990). Under surplus ($[G]^+ = 0$), the sensing term vanishes and the firm accumulates no capability from successful operations. A surplus-learning extension (adding $r'[\text{-}G]^+W$ with $r' \ll r$) would not qualitatively alter the results because the Front-Stage Trap, Good Jobs Buffer, and Burnout Cliff are driven by overload-regime feedback; surplus-regime learning would modestly raise K^* at the healthy equilibrium without affecting the instability thresholds.

Workforce well-being. Well-being rises under surplus capacity and direct investment; it falls through natural decay and overload-amplified depletion:

$$\frac{dW}{dt} = \big([-G]^+ + f\big)(1-W) - \big(\mu + \sigma\,[G]^+\big)W. \tag{6}$$

Two grouped terms ensure $W(t) \in [0,1]$. The first drives W upward (vanishing at $W = 1$): capability surplus and direct workforce investment raise well-being with diminishing returns, consistent with the JD-R model

(Bakker and Demerouti 2007). The second drives W downward (vanishing at $W = 0$): natural decay (μ) and overload erosion ($\sigma\,[G]^+$) deplete well-being, with $\sigma > 1$ reflecting the asymmetry documented by Oliva and Sterman (2001). All three theorems hold for any $\sigma > 1$ (EC Table EC.3).

Customer trust. Trust builds through reliable service and erodes faster when reliability fails (Sirdeshmukh et al. 2002, Mittal et al. 1998):

$$\frac{dT}{dt} = \beta\,[-G]^+\,(1-T) \;-\; \eta\,\beta\,[G]^+\,T \;-\; \lambda T, \tag{7}$$

where $\beta > 0$ is the trust-update rate, $\eta > 1$ is the erosion asymmetry, and $\lambda > 0$ is natural trust decay. We conservatively set $\eta = 2.0$, below the approximately 2.5:1 depletion-to-enhancement ratio reported by Sirdeshmukh et al. (2002).

The system (5)–(7) is piecewise-smooth; existence and uniqueness of solutions follow from Filippov's (1988) framework (details and smoothing robustness in EC §EC.1.1 and EC.1.8).

3.4. State-Space Invariance and Empirical Mapping

The state space $K > 0, W \in [0,1], T \in [0,1]$ is forward-invariant for all $t \geq 0$; the verification via directional derivatives at each boundary is given in EC §EC.1.1.

Parameter calibration details are in EC §EC.4.

3.5. Feedback-Loop Architecture

Figure 1 shows the causal-loop architecture of the three ODEs, using the conventions of Sterman (2000). The diagram contains four feedback loops, each traceable from the state-variable arrows.

Loop R1 ($T \to D \to G \to T$): regime-dependent polarity; balancing when capacity exceeds demand (surplus absorbs trust-driven demand growth), but reinforcing under overload (trust loss reduces demand less than capacity collapses, widening the gap). **Loop R2** ($G \to W \to K_{\text{eff}} \to G$, reinforcing): the deterioration loop behind the Front-Stage Trap. Absent from prior quality-erosion models because it requires the capacity multiplier $h(W)$. **Loop R3** ($W \to K_{\text{eff}} \to G \to W$, reinforcing): the escalation channel behind the Burnout Cliff, likewise novel to this model. **Loop B1** ($G \to K \to K_{\text{eff}} \to G$, balancing): sensing investment closes the gap, but only if workforce capacity W is sufficient to support adaptation (see $r[G]^+W$ in Eq. 5). Loops R2 and R3, created by the capacity multiplier $h(W)$, have no counterpart in prior quality-erosion models; they are the source of the regime-dependent instability that drives all three theorems.

Figure 1 **Causal-loop architecture. Solid boxes are state variables; dashed boxes are exogenous controls. Arrow signs indicate direction of effect; loop polarity is shown in parentheses. R2 and R3, created by the capacity multiplier $h(W)$, are novel and do not exist in single-domain models.**

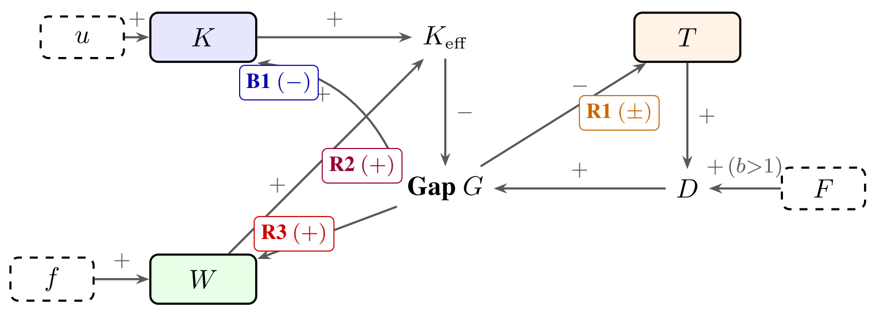


Table 2 **Model parameters, baseline values, and kill conditions.**

#	Parameter	Baseline	Timescale	Kill	Mechanism killed	Empirical anchor
1	b (demand super-linearity)	1.3	–	b=1	Front-Stage Trap	Bell et al. (2018); Acimovic & Graves (2015)
2	a (trust-demand coupling)	0.4	–	a=0	Demand feedback	McKnight et al. (2002); Sirdeshmukh et al. (2002)
3	p (workforce complementarity)	0.5	–	p=0	Capacity multiplier	Krusell et al. (2000)
4	σ (overload asymmetry)	2.0	–	σ=1	Tipping point	Oliva and Sterman (2001); Bakker and Demerouti (2007)†
5	η (erosion asymmetry)	2.0	–	η=1	Trust asymmetry	Sirdeshmukh et al. (2002)
6	r (sensing-adaptation)	0.4	$\tau_r \approx 1.7$ mo	r=0	Self-correction dies	Tucker (2007); Argote & Epple (1990)
7	δ (depreciation)	0.03	$\tau_\delta \approx 23$ mo	–	–	Illustrative (EC §EC.4)
8	β (trust-update rate)	0.12	$\tau_\beta \approx 5.8$ mo	$\beta \to 0$	Trust frozen	Illustrative (EC §EC.4)
+	μ (natural W decay)	0.05	$\tau_\mu \approx 14$ mo	μ=0	W experiential only	Demerouti et al. (2001); Dubé et al. (2020)
+	λ (natural T decay)	0.025	$\tau_\lambda \approx 28$ mo	λ=0	T experiential only	Illustrative (EC §EC.4)

All time units are months. Timescales are $\ln(2)/\theta$ (half-life). Eight free parameters (rows 1–8); μ and λ govern natural decay. "Illustrative" parameters are calibrated within empirically informed ranges (EC §EC.4). †Magnitude comparator from service industry evidence; see §3.3. Direct estimation of b and p from retailer panel data is the primary empirical extension (§7.2).

3.6. Parameters and Special Cases

Table 2 summarises all parameters. Two control variables complete the model: $u \geq 0$ (capability investment) and $f \geq 0$ (workforce investment). The model recovers the core dynamics of Oliva and Sterman (2001) when $p = 0$ and $a = 0$ (see EC §EC.2 for the structural mapping), and reduces to a Solow-type capital-accumulation equation when $p = 0$, $a = 0$, and $r = 0$.

4. Analytical Results

We derive four formal results: Theorems 1–3 characterise three phenomena that no single-domain model can produce; Proposition 4 establishes digitalisation-workforce complementarity. Full proofs are in the Electronic Companion (§EC.1); here we provide proof sketches.

4.1. Equilibrium Structure

The model generates two distinct long-run outcomes. The *surplus equilibrium*: effective capability exceeds demand, the workforce is healthy, trust builds, and the firm settles into a self-reinforcing high-performance state. The *low-trust attractor*: demand has overwhelmed capability, the workforce is depleted, and customers have withdrawn trust; exit requires structural intervention.

The technical characterisation of these two states is as follows. The surplus equilibrium exists for all $F < F_{\text{crit}}$ and is unique under Condition C2 (EC Lemma 2, §EC.1.2). When effective capability exceeds demand, define the operational surplus $S^* \equiv K^* h(W^*) - D(T^*, F) \geq 0$ as the excess of effective capability over demand at equilibrium. Equilibrium trust is an increasing function of S^*, saturating toward one as surplus grows: $T^* = \beta S^*/(\beta S^* + \lambda)$. Equilibrium well-being is similarly increasing in surplus and investment: $W^* = (S^* + f)/(S^* + f + \mu)$. The capability stock settles at $K^* = u/\delta$, determined entirely by the investment-depreciation balance. Full algebraic derivations are in EC §EC.1.2. Both T^* and W^* are increasing in S^*, so any improvement in effective capability propagates into higher trust and well-being at the equilibrium, creating the reinforcing dynamics that underpin the Good Jobs Buffer.

The low-trust attractor is reached whenever demand persistently overwhelms capability. With $T^* \approx 0$, demand collapses to its digitalisation-driven floor and the workforce settles at a low but strictly positive well-being level, held above zero by the workforce investment rate f. Effective capability at this attractor falls short of demand, maintaining a persistent residual gap. The attractor exists whenever $F^b > (u/\delta)\, h(W^*_{\text{low}})$ and is locally stable (EC Remark 3).

Maintained conditions. The formal results hold under four regularity conditions (C1)–(C4), verified analytically or numerically across the parameter space (EC Remark 5): surplus equilibrium existence, uniqueness, regularity, and low-trust attractor existence (EC §EC.1.2). The coexistence of surplus equilibrium and low-trust attractor constitutes bistability; the transition at F_{crit} where these two states coalesce is classified as a saddle-node (fold) bifurcation (EC Remark 5).

REMARK 1 (ANALYTICAL TRANSPARENCY). The table below classifies each result by evidentiary basis: Tier A (analytic for all valid parameters), Tier B (analytic under C1–C4), or Tier C (computational verification).

Result	**Tier**	**Basis**
Thm 1(i): equilibrium existence/nonexistence	A	Analytic (IVT + $\Phi(0)$ sign)
Thm 1(ii): trajectories bounded	A	Analytic (Gronwall; Lemma 3)
Thm 1(ii): wide-basin convergence	C	Numerical grid, $K_0 = K^*$ slice
Lemma 2 (uniqueness of S^*)	B	Analytic under C2-SC ($\beta(f+\mu) < \lambda$; holds throughout parameter space; EC §EC.1.2)
Thm 2: $\partial\,\text{CLV}^*/\partial f > 0$ (sign)	B	IFT chain; GE feedback correction sign analytic
Thm 2: GE feedback magnitude (≈0.09)	C	Confirmed across 100 configurations
Thm 3(i): $\Delta F_{\text{collapse}}$ threshold	A	Direct from Thm 1
Thm 3(i): convergence from pre-shock IC	C	Numerical; basin-dependent
Prop 4 cross-partial	A	Analytic ($K_{\text{eff}} = Kh(W)$)
Prop 4 $F_{\max}(W)$ boundary formula	A	Exact at $S^* \to 0$
Prop 4 $F_{\max}(W)$ interior upper bound	B	Analytic sign; ≈8% correction numeric
Corollary 5, Case 1	B	Thm 1 + Thm 2 directly
Corollary 5, Case 2 near $B_{\min}$	B	Continuity from Case 1 (IFT; EC §EC.1.7)
Corollary 5, Case 2 interior	C	Computational; $c_F/c_f \in [0.5, 5]$

4.2. Theorem 1: The Front-Stage Trap

THEOREM 1 **(Front-Stage Trap).** *Suppose $b > 1$ and $p > 0$. Define the digitalisation threshold*

$$F_{\text{crit}} = \left(\frac{u}{\delta} \cdot \left[(1-p) + p\,\frac{f}{f+\mu}\right]\right)^{1/b}, \tag{8}$$

which depends only on exogenous parameters $(u, \delta, p, f, \mu, b)$ and equals $K^* h(f/(f+\mu))^{1/b}$, the capacity multiplier evaluated at the boundary workforce level. Then:

(i) For $F < F_{\mathrm{crit}}$, the surplus equilibrium (unique under C2; Lemma 2, EC §EC.1.2) exists with $T^* > 0$ and $\mathrm{CLV}^* > \mathrm{CLV}_{\mathrm{low}}$[1] (the CLV at the low-trust attractor).

(ii) For $F \geq F_{\mathrm{crit}}$, no surplus equilibrium exists. All forward trajectories remain in a bounded compact set (Lemma 3, EC §EC.1.2), and the low-trust attractor is locally stable (EC Remark 3). Wide-basin convergence to this attractor is confirmed numerically (Tier C; Remark 1).

The threshold F_{crit} is decreasing in p and increasing in f: the Trap binds earlier when workforce-technology coupling is strong, and is relaxed by workforce investment.

Proof sketch. The surplus equilibrium requires $\Phi(0) > 0$ (where $\Phi(S) \equiv K^* h(W^*(S)) - D(T^*(S), F) - S$ is the surplus residual; full construction in EC §EC.1.2), which holds if and only if $F < F_{\mathrm{crit}}$. At $F = F_{\mathrm{crit}}$, the surplus $S^* = 0$ and the equilibrium ceases to exist. For $F > F_{\mathrm{crit}}$, $G > 0$, activating overload depletion of W through Eq. (6), which reduces $h(W)$, which further widens G through Loop R2. The reinforcing dynamics drive the system toward the low-trust attractor. Boundedness follows from Lemma 3 (EC §EC.1.2): $\dot{K} \leq C(F) - \delta K$ by Gronwall's lemma, and $W, T \in [0, 1]$ by construction. Full proof in EC §EC.1.3. □

At the calibrated baseline, $F_{\mathrm{crit}} \approx 1.11$, corresponding to roughly a 40% increase in digital feature complexity above $F_0 = 0.80$. Figure 2 maps the regime structure: green cells converge to the surplus attractor (high CLV); red cells converge to the low-trust attractor (CLV ≈ 400). The cross-domain CLV sign reversal requires $b > 1$ (super-linear demand) and $p > 0$ (workforce-contingent threshold); at $p = 0$ the threshold reduces to $(u/\delta)^{1/b}$ and no longer responds to workforce investment.

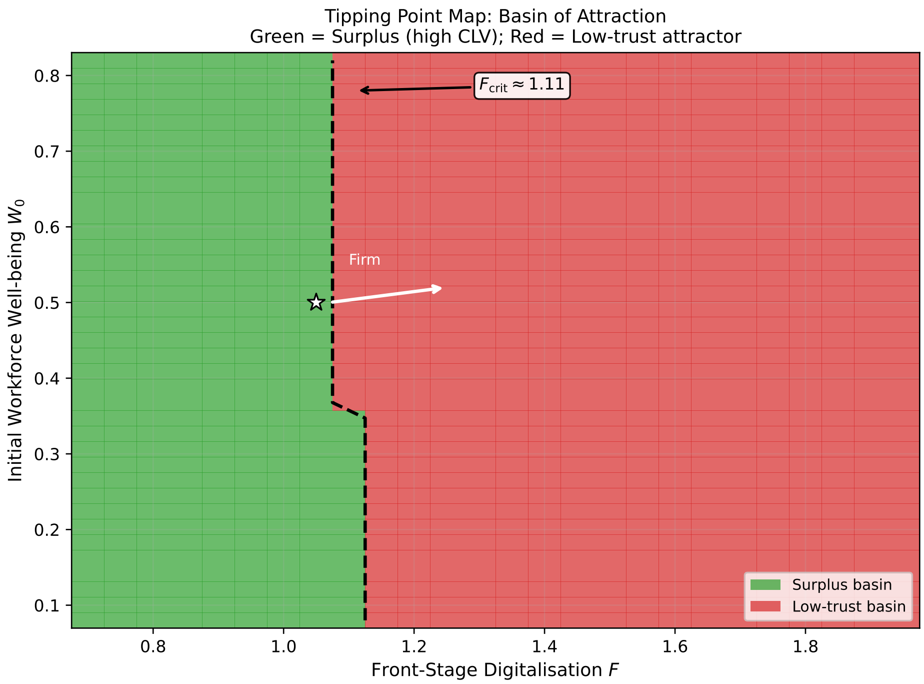
4.3. Theorem 2: The Good Jobs Buffer

THEOREM 2 **(Good Jobs Buffer).** *Suppose $p > 0$ and the surplus equilibrium exists and is stable (Conditions C1–C4, i.e., the surplus equilibrium exists, is unique, and the low-trust attractor is present; Remark 1). Workforce investment f raises steady-state CLV through a multiplicative capacity-level channel:*

$$\frac{\partial\, \mathrm{CLV}^*}{\partial f} > 0 \qquad \textit{for all } f \geq 0. \tag{9}$$

[1] CLV is formally calibrated in §5; the theorems require only that CLV is increasing in T^*, which holds for any standard customer-lifetime-value functional.

Figure 2 **Basin-of-attraction map. Green cells converge to the surplus attractor (high CLV) for $F < F_{\mathrm{crit}} \approx 1.11$; red cells converge to the low-trust attractor for $F \geq F_{\mathrm{crit}}$. Doubled-grid validation confirms boundary stability (EC §EC.4.2).**



The return decomposes into a direct capacity effect and an indirect trust effect:

$$\frac{\partial\,\mathrm{CLV}^*}{\partial f} = \underbrace{\frac{\partial\,\mathrm{CLV}}{\partial T^*} \cdot \frac{\partial T^*}{\partial S^*} \cdot K^* \cdot p}_{\textit{direct: } h(W) \textit{ raises } K_{\mathrm{eff}}} \cdot \frac{\partial W^*}{\partial f}$$

$$+ \underbrace{\textit{GE trust-demand feedback}}_{\textit{indirect: } T^* \to D\textit{, dampened by } \approx 0.09\textit{; EC §EC.1.4}} .$$

Proof sketch. Since $W^* = (S^* + f)/(S^* + f + \mu)$ at the surplus equilibrium, differentiating gives $\partial W^*/\partial f = \mu/(S^* + f + \mu)^2 > 0$: workforce investment raises steady-state well-being. Higher W^* raises $h(W^*) = (1 - p) + pW^*$, which increases $K_{\mathrm{eff}} = K^* \cdot h(W^*)$, which widens the surplus S^*, which raises $T^* = \beta S^*/(\beta S^* + \lambda)$, which raises CLV. Each link is strictly positive: $\partial h/\partial W = p > 0$, $\partial S/\partial h = K^* > 0$, $\partial T^*/\partial S^* > 0$, $\partial\,\mathrm{CLV}/\partial T^* > 0$. The chain-rule product is therefore positive, and diminishing returns follow from the concavity of W^* in f. Full proof in EC §EC.1.4. $\square$

The p=0 counterfactual in §5.1 confirms this: disabling the workforce multiplier leaves equilibrium CLV exactly invariant to workforce spending. The entire CLV return flows through service capacity, not the operational sensing channel: the indirect sensing channel contributes less than 1% of the total CLV uplift across all tested p values (EC Table EC.4). Standard dashboards, which record technology and workforce returns in separate ledgers, structurally underestimate the financial return to workforce investment.

4.4. Theorem 3: The Burnout Cliff

THEOREM 3 **(Burnout Cliff).** *Suppose $\sigma > 1$, $p > 0$, and $F_0 < F_{\text{crit}}$ (a surplus equilibrium exists at F_0). Under these conditions the system is bistable in a neighbourhood of F_{crit} (EC Remark 5). Consider a firm at the surplus equilibrium that implements a step increase in F from F_0 to $F_0 + \Delta F$. Two thresholds govern the outcome:*

(i) Permanent-collapse onset. *For $\Delta F \geq \Delta F_{\text{collapse}}$, no surplus equilibrium exists at the post-upgrade level. The low-trust attractor is the only locally stable state (EC Remark 3); convergence to it is confirmed numerically across the studied initial conditions.*

$$\begin{aligned}\Delta F_{\text{collapse}} &= F_{\text{crit}} - F_0 \\ &= \left(\frac{u}{\delta}\left[(1-p) + p\tfrac{f}{f+\mu}\right]\right)^{1/b} - F_0.\end{aligned}$$

For $\Delta F < \Delta F_{\text{collapse}}$, a new surplus equilibrium exists at the post-upgrade level. Whether a given trajectory reaches this equilibrium depends on its initial condition relative to the basin boundary (Figure 2); numerical verification confirms convergence from the pre-shock equilibrium initial condition across the parameter configurations studied (EC §EC.4.2).

(ii) Instantaneous-overload diagnostic. *A second, larger threshold marks the boundary above which the service gap turns positive at $t = 0^+$:*

$$\Delta F_{\text{rapid}} = \frac{S_0}{b \cdot F_0^{b-1} \cdot (1 + aT^*)}, \tag{10}$$

where $S_0 = K^* h(W^*) - D(T^*, F_0)$ is the pre-upgrade surplus. Because $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ (at baseline: 0.40 vs. 0.31), a firm in the zone $\Delta F \in [\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ collapses permanently but through a slow *trajectory (G negative at $t = 0^+$, surplus eroded gradually). For $\Delta F \geq \Delta F_{\text{rapid}}$, G turns positive immediately and the system enters a rapid self-reinforcing collapse to the low-trust attractor.*

The safe zone is $\Delta F < \Delta F_{\text{collapse}}$; ΔF_{rapid} is a deployment-speed diagnostic, not a safety threshold.

Proof sketch. Part (i) follows directly from Theorem 1: for $F_0 + \Delta F \geq F_{\text{crit}}$, no surplus equilibrium exists; for $\Delta F < \Delta F_{\text{collapse}}$, a new surplus equilibrium exists at $F_0 + \Delta F$, and numerical verification confirms that trajectories originating at the pre-shock surplus equilibrium converge to this new surplus equilibrium across the parameter configurations studied (EC §EC.4.2). For Part (ii), setting the first-order demand increment $\Delta D \approx (1 + aT^*)\, b\, F_0^{b-1}\, \Delta F$ equal to the pre-upgrade surplus S_0 gives ΔF_{rapid}. When $\Delta F < \Delta F_{\text{rapid}}$, $G_0^+ = 0$ at t_0^+ and the system crosses the workforce tipping threshold through a slow transient, consistent with the slow-collapse zone. Full proof in EC §EC.1.5. □

Workforce investment f raises F_{crit} and enlarges the safe upgrade zone. Before each deployment phase, the firm should compute $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ as a hard ceiling on planned upgrade size; this ceiling is set entirely by f.

4.5. Proposition 4: Digitalisation-Workforce Complementarity

PROPOSITION 4 (**Digitalisation-Workforce Complementarity**). *The marginal value of workforce improvement is strictly increasing in the technology base:*

$$\frac{\partial K_{\text{eff}}}{\partial W} = p \cdot K, \qquad \frac{\partial^2 K_{\text{eff}}}{\partial W\, \partial K} = p > 0. \tag{11}$$

Since maintaining surplus at higher F requires higher effective capability K_{eff} (that is, either higher K^* through increased u, or higher $h(W^*)$ through increased f), the equilibrium return to workforce investment increases with digitalisation level.

Equivalently, workforce well-being determines the maximum sustainable digitalisation level. At the regime boundary ($S^* \to 0$), the equilibrium trust approaches zero ($T^* \to 0$; see Lemma 2, EC §EC.1.2), so:

$$F_{\max}(W) \;=\; \big(K^* \cdot h(W)\big)^{1/b}, \tag{12}$$

which is strictly increasing in W for $p > 0$. The full implicit differentiation accounting for $T^*(W, F)$ dependence at interior equilibria (where $S^* > 0$) yields a correction term of order $O(aS^*)$ that vanishes as $S^* \to 0$; the formula (12) is therefore exact at the boundary and an upper bound on $F_{\max}$ in the interior: trust-elevated demand means true safe capacity is approximately 8% lower than this formula implies at baseline trust levels (EC §EC.1.6).

Proof. The cross-partial follows directly from $K_{\text{eff}} = K[(1-p) + pW]$. The $F_{\max}$ formula identifies F_{crit} at $S^* \to 0$, where $T^* \to 0$, giving $(K^* h(W))^{1/b}$, strictly increasing in W since $h'(W) = p > 0$; with the full $T^*(W, F)$ correction this is exact at the boundary and an upper bound in the interior (EC §EC.1.6). □

Remark. The constant cross-partial $\partial^2 K_{\text{eff}}/\partial W\,\partial K = p$ is an immediate consequence of the linear specification $h(W) = (1-p) + pW$. Under a concave alternative $h(W) = (1-p) + pW^\gamma$ ($\gamma < 1$), the cross-partial becomes $p\gamma W^{\gamma-1}$, which is state-dependent: complementarity is stronger when the workforce is depleted (low W), precisely when it matters most operationally. The linear form therefore understates the complementarity effect under overload, making our result a conservative bound. Section 7.2 discusses additional implications of concave h.

Figure 3 illustrates: the gap between a stressed firm ($W = 0.30$, $F_{\max} \approx 1.063$) and a healthy one ($W = 0.70$, $F_{\max} \approx 1.307$) is 23%, arising entirely from workforce investment.

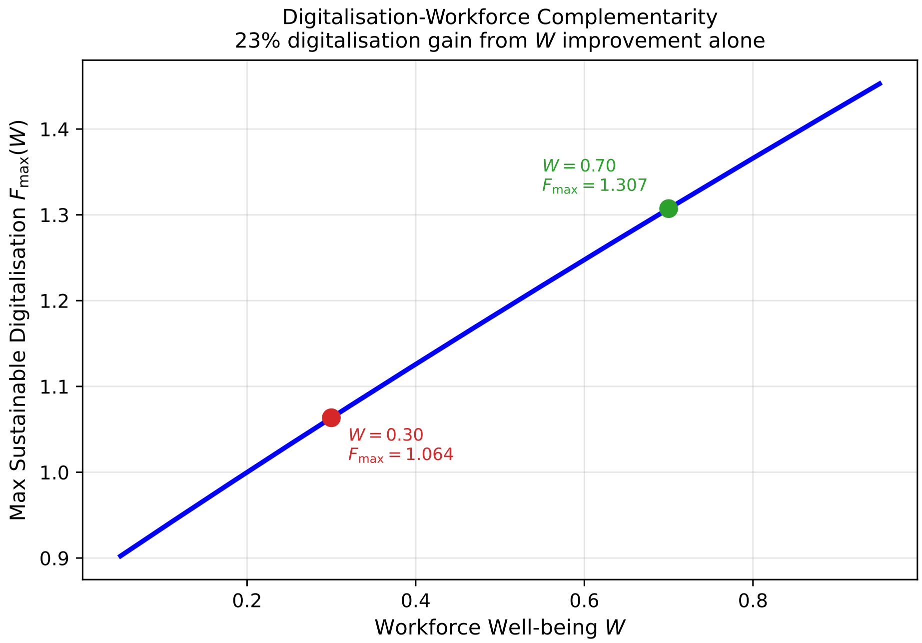
COROLLARY 5 **(Budget-Constrained Trap).** *Under a fixed budget constraint $c_F F + c_f f \leq B$ with $B \geq B_{\min} - \varepsilon$ (where $B_{\min} = c_F \cdot F_{\text{crit}}$), the CLV-maximising allocation includes positive workforce investment $f^* > 0$: the all-F corner is strictly dominated analytically whenever the budget reaches or exceeds the Trap threshold, and computationally for B throughout $[B_{\min}/2,\, B_{\min})$ across cost ratios $c_F/c_f \in [0.5, 5]$.*

Proof. See EC §EC.1.7. □

5. Computational Study

We illustrate the analytical results through calibrated simulations using the baseline parameters in Table 2. CLV follows Gupta et al. (2004): $\text{CLV} = w(T)/(1-\rho(T))$, where $\rho(T)$ is an annual retention probability and $w(T)$ an annualised per-customer margin, with $\rho(T) = 0.80 + 0.18T$ and $w(T) = 80(1 + 0.25\max(T -$

Figure 3 Digitalisation-workforce complementarity. The healthy firm ($W = 0.70$) sustains 23% higher $F_{\max}$ than the stressed firm ($W = 0.30$), arising entirely from workforce investment.



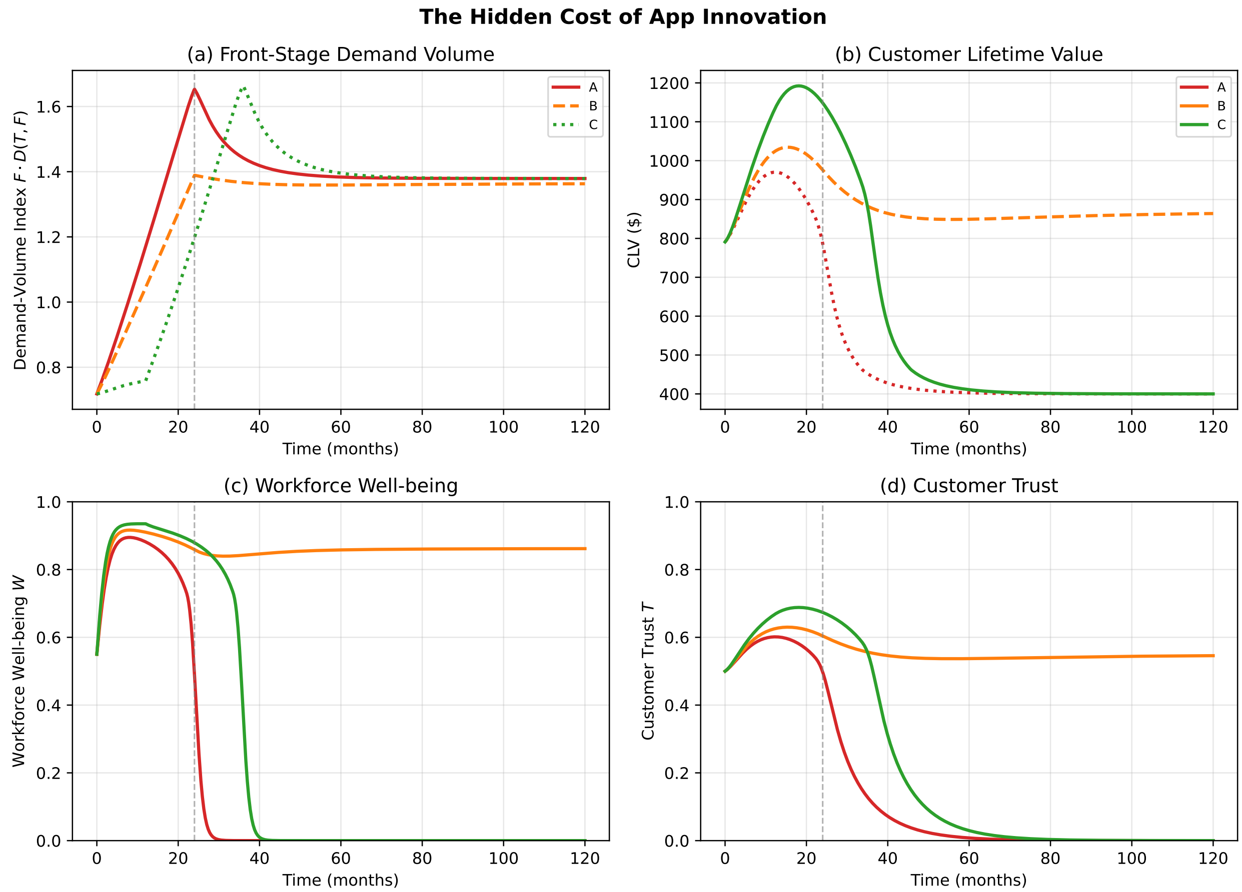
$0.15, 0)$). The endpoints $\rho(0) = 0.80$ and $\rho(1) = 0.98$ fall within the 70–90% annual range in Gupta et al. (2004); \$80 is an illustrative annual margin for US omnichannel grocery. All analytical results hold for any CLV increasing in trust (calibration in EC §EC.4.3). All three qualitative phenomena are robust across $b \in [1.1, 2.0]$, $\sigma \in (1.0, 3.0]$, and $p \in [0.25, 0.75]$ (EC Table EC.3; at $\sigma = 1$ the Burnout Cliff disappears).

5.1. The Hidden Cost of App Innovation (Figure 4)

Three deployment strategies span the critical threshold $F_{\text{crit}} \approx 1.11$. All begin from $(K_0, W_0, T_0) = (1.60, 0.55, 0.50)$ with $F_0 = 0.80$. Scenario A ramps aggressively to $F_{\text{end}} = 1.15$ with no workforce investment; Scenario B ramps to $F_{\text{end}} = 1.05$ with concurrent support ($f = 0.06$); Scenario C pre-invests in workforce ($f = 0.10$, months 1–12, after which f returns to 0.0) before ramping to $F_{\text{end}} = 1.15$.

All three appear profitable on front-stage metrics (panel a), yet CLV diverges sharply (panel b): by month 24, Scenario C sustains CLV approximately 40% above Scenario A. Workforce collapse precedes trust collapse (panels c, d), confirming the cross-domain cost invisible in the technology business case.

Figure 4 **The hidden cost of app innovation. Panel (a): the demand-volume index $F \cdot D(T(t), F(t))$ rises for all three scenarios, masking divergent back-stage outcomes. Panel (b): CLV diverges sharply by month 24. Panels (c,d): workforce collapse precedes trust collapse in the no-investment scenario.**



5.2. Basin-of-Attraction Structure

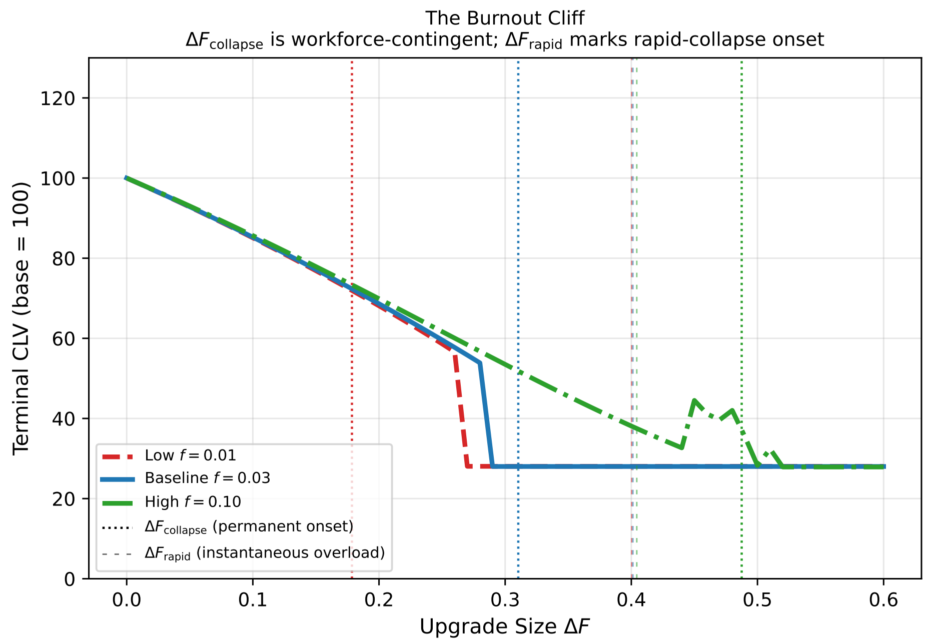
Figure 2 maps the basin boundary across the joint space of W_0 and F. The boundary shifts rightward as W_0 increases: a firm near the basin boundary can increase its safety margin by raising workforce investment f, which shifts F_{crit} rightward without touching the technology stock.

5.3. The Burnout Cliff (Figure 5)

We sweep $\Delta F \in [0, 0.60]$ for three workforce-investment scenarios: neglected ($f = 0.01$), baseline ($f = 0.03$), and invested ($f = 0.10$), each starting from its surplus equilibrium at $F_0 = 0.80$. Varying f shifts $F_{\text{crit}}(f)$ from 0.98 to 1.29, widening $\Delta F_{\text{collapse}}$ from 0.18 to 0.49.

Figure 5 shows the three normalised CLV curves. For the neglected firm ($f = 0.01$), the safe zone collapses to $\Delta F < 0.18$; for the invested firm ($f = 0.10$), it extends to $\Delta F = 0.49$. The slow-collapse zone between

Figure 5 **The Burnout Cliff. Normalised terminal CLV versus upgrade size ΔF for three workforce-investment levels. Dotted verticals: $\Delta F_{\text{collapse}}$ (0.18, 0.31, 0.49). The neglected firm ($f = 0.01$) exhausts its safe zone at $\Delta F = 0.18$; the invested firm ($f = 0.10$) retains headroom to $\Delta F = 0.49$.**



$\Delta F_{\text{collapse}}$ and ΔF_{rapid} is the diagnostic blind spot: dashboards report continued growth while the trust trajectory has already turned negative. For the well-investing firm, this zone disappears entirely (EC §EC.1.5).

6. Empirical Evidence

We examine three observational patterns implied by the model's predictions, using complementary identification strategies at different levels of aggregation: (i) a firm-level inverted-U test and event study using customer satisfaction data ($N = 216$ for the quadratic specification, 17 firms; $N = 108$ for the event study, 16 firms[2]); (ii) a county-level panel test using Bureau of Labor Statistics (BLS) retail employment and wages ($N = 30{,}951$, 3,115 US counties). The firm-level analysis tests the Front-Stage Trap (inverted-U and post-shock ACSI decline). A cross-sectional complementarity check (Glassdoor $\times$ technology interaction) provides a directional sign check but lacks statistical power ($N_c = 13$). The county-level analysis tests the Good Jobs Buffer at a different margin: whether counties with higher initial retail wages show more resilient

[2] Amazon is excluded from the event study because its omnichannel transition was gradual and continuous rather than marked by a discrete launch event, making event-time identification infeasible.

Table 3 Model-to-data mapping.

Model variable	Empirical proxy	Justification
F (digital depth)	E-commerce share (FRED ECOMPCTSA)	National digital penetration index
W (workforce well-being)	Glassdoor overall rating (1–5)	Composite employee engagement
T (customer trust)	ACSI score (0–100)	Aggregate customer confidence
K (technology stock)	CapEx / Revenue	Back-stage capital intensity

retail employment as e-commerce penetration grows. An independent European sample ($N_c = 11$, $N = 20$) is reported in the EC (§EC.5) as an exploratory out-of-sample check.

6.1. Data and Samples

Table 3 maps each model variable to its empirical proxy.

Firm-level sample. We assemble an annual panel of 17 publicly listed US retailers from 2008 to 2024 using SEC EDGAR 10-K filings. Customer satisfaction (T proxy) is measured by the American Customer Satisfaction Index (ACSI; Fornell et al. 1996; $N = 216$, 17 firms). Workforce well-being (W proxy) uses Glassdoor overall ratings (1–5 scale) (Corbett 2024). Technology investment (K proxy) is capital expenditure as a share of revenue. E-commerce share is from the Federal Reserve (FRED series ECOMPCTSA). The inverted-U specification requires only ACSI and e-commerce ($N = 216$, 17 firms); the complementarity specification additionally requires Glassdoor and CapEx, yielding 131 complete firm–year observations across 13 firms after listwise deletion.

County-level sample. The BLS Quarterly Census of Employment and Wages provides county-level annual average retail employment, weekly wages, and establishment counts for the retail trade sector (NAICS 44–45), 2015–2024. We retain counties with baseline retail employment exceeding 50 and weekly wage exceeding $200, yielding 30,951 county–year observations across 3,115 counties.

Table 4 reports descriptive statistics for both samples.

6.2. Firm-Level Results

Cross-sectional complementarity (Proposition 4). The between-effects (OLS) estimate of the standardised interaction $K^z \times W^z$ is positive ($+1.95$, $p = 0.41$; $N = 131$, 13 firms). The sign is consistent with the prediction that firms with jointly higher technology and workforce investment enjoy higher satisfaction, and aligns directionally with Aral et al. (2012). However, the coefficient is not statistically significant, reflecting

Table 4 Descriptive statistics.

Variable	N	Mean	SD	Range
Panel A: Firm-level sample (13 firms, 2010–2024)				
ACSI score	131	76.5	3.4	68–88
Glassdoor rating (1–5)	131	3.23	0.30	2.7–3.9
CapEx / Revenue (%)	131	2.9 [2.0–4.5]†		
Panel B: County-level sample (3,115 counties, 2015–2024)				
Annual avg. retail employment	30,951	5,035	15,458	18–423,900
Annual avg. weekly wage ($)	30,951	558	151	203–2,861
Baseline wage, 2015 ($)	3,115	476	110	203–1,697
Baseline employment, 2015	3,115	5,053	15,712	51–416,188
Log employment growth (vs. 2015)	30,951	0.00	0.12	-1.5–2.9
National e-commerce share (%)	10	11.8	3.0	7.4–16.1

†CapEx/Revenue reports the median [IQR] rather than the mean and SD due to extreme outliers in the raw ratio (e.g., BBY 2012). Panel A describes the 13-firm subsample with joint ACSI, Glassdoor, and CapEx coverage (used for the complementarity test). The inverted-U and event-study specifications use a broader 17-firm and 16-firm ACSI-only sample ($N = 216$ and $N = 108$, respectively); descriptive statistics for these samples are available upon request.

the extreme power limitation of 13 firm clusters (Cameron et al. 2008). We treat this as a sign check, not a test. Within-firm estimates (firm FE) are negative, as expected given the low within-firm Glassdoor SD (30% of total SD); the between-effects estimate is the appropriate estimand for a structural cross-firm relationship (Milgrom and Roberts 1990).

Inverted-U (Theorem 1). We estimate the quadratic specification

$$\mathrm{ACSI}_{it} = \alpha_i + \beta_1\,\mathrm{Ecom}_t + \beta_2\,\mathrm{Ecom}_t^2 + \mathbf{X}_{it}\theta + \varepsilon_{it}, \tag{13}$$

where α_i are firm fixed effects, Ecom_t is national e-commerce share, and $\mathbf{X}_{it}$ includes workforce controls where indicated. The specification ($N = 216$, 17 firms) yields $\hat{\beta}_2 = -0.028$ ($p = 0.028$ under $t(16)$), consistent with the predicted inverted-U. The estimated turning point (7.9% e-commerce share) lies in the mid-2010s, qualitatively consistent with the model's prediction of a digitalisation ceiling, though we caution that the turning point is on the e-commerce share scale while F_{crit} is a dimensionless technology index; the correspondence is directional, not quantitative. The coefficient attenuates when workforce controls are included ($p = 0.66$), consistent with the model's prediction that e-commerce damages satisfaction *through* workforce depletion. A limitation common to both the inverted-U and complementarity specifications is the absence of instruments: cross-sectional OLS cannot exclude the possibility that unobserved management quality drives both higher Glassdoor ratings and higher ACSI (Ton 2014). The event study partially mitigates this concern by exploiting within-firm temporal variation around exogenous technology shocks.

Event study (Theorem 1). We exploit the staggered timing of major omnichannel launches (e.g., Walmart curbside 2017, Target Drive Up 2018, Kroger ClickList 2016) as firm-level technology shocks. Using a $[-3,+3]$ year event window around each firm's launch ($N = 108$, 16 firms), a two-way FE specification with pre-shock capital expenditure as control yields a post-shock ACSI decline of -0.71 ($p = 0.037$ under $t(15)$), consistent with the Front-Stage Trap: major omnichannel deployments reduce customer satisfaction on average. Event-time coefficients for the level specification show near-zero and insignificant pre-trend coefficients at $t = -3$ and $t = -2$ (relative to $t = -1$), supporting the parallel-trends assumption for the basic difference-in-differences (EC Table EC.7). Pre-trend coefficients for the workforce-moderation interaction are similarly insignificant (EC Table EC.7). The workforce moderation (post $\times$ pre-Glassdoor) is not detectable ($p = 0.79$), consistent with the low within-firm Glassdoor variation rather than absence of the mechanism.

6.3. County-Level Good Jobs Buffer

The firm-level data cannot deliver a well-powered test of the capacity-multiplier channel. We therefore test an *analogue* of the Good Jobs Buffer at a different margin (employment resilience rather than customer satisfaction) using BLS county-level data, where the dramatically larger sample ($N = 30{,}951$) provides the statistical power that the firm-level analysis lacks.

Design. The model predicts that workforce investment f buffers the negative effects of digitalisation on service capacity (Theorem 2). The county-level test maps this prediction to an observable analogue: do counties with higher initial retail wages (a proxy for local workforce-investment conditions) show more resilient retail employment as national e-commerce penetration grows? Three mapping choices warrant explicit acknowledgement. First, the outcome is employment resilience, not customer lifetime value; the county test identifies a labour-market analogue of the capacity-multiplier channel rather than the customer-satisfaction channel central to Theorem 2. Second, county-level retail wages reflect local labour-market conditions (including minimum-wage floors, union density, and regional cost of living) beyond any individual firm's workforce investment f; they are a noisy proxy for the model's control variable. Third, national e-commerce share Ecom_t is a macro trend common to all counties, whereas the model's digitalisation level F is a firm-level choice. Despite these distinctions, the qualitative prediction is preserved: higher workforce investment should buffer against digitalisation-induced capacity loss, and the county-level test provides a well-powered setting in which to evaluate this prediction. Initial retail wage is measured in 2015 (before the e-commerce acceleration), standardised cross-sectionally. The interaction $W^z_{c,2015} \times \text{Ecom}_t$ captures whether high-wage counties differentially maintain employment under digital pressure:

$$\ln E_{ct} - \ln E_{c,2015} = \alpha_c + \gamma_t \tag{14}$$

$$+\beta_1 \left(W^z_{c,2015} \times \text{Ecom}_t\right) + \varepsilon_{ct}. \tag{15}$$

Because Ecom_t is a national-level variable (identical across counties within each year), its main effect is perfectly collinear with the year fixed effects γ_t and is therefore absorbed by them. Likewise, $W^z_{c,2015}$ is time-invariant and absorbed by the county fixed effects α_c. The coefficient β_1 thus identifies the *differential* effect of e-commerce growth on counties with higher initial retail wages, which is the quantity of interest for the Good Jobs Buffer.

Results. The interaction is positive and significant ($\hat{\beta}_1 = 0.0026$, $t = 2.24$, $p = 0.025$; county and year FE, clustered at county). A county one SD above the mean in initial retail wage (≈\$110/week) retains 2.6 percentage points more retail employment per 10-point increase in e-commerce share, consistent with the Good Jobs Buffer operating through employment capacity. The long-difference specification (2015 vs. 2023) confirms: high-wage counties gained 2.1 pp more retail employment over the period ($t = 2.57$, $p = 0.010$).

Robustness. The result strengthens when COVID years are excluded ($\hat{\beta}_1 = 0.0031$, $p = 0.014$), holds in urban counties ($p = 0.006$), and is insignificant in rural counties ($p = 0.58$). The urban–rural difference is consistent with the mechanism operating where retail employment is concentrated, though it could also reflect that the wage proxy captures urban labour-market dynamics (cost of living, competition) rather than

Table 5 Firm-level regression results.

	Complementarity		*Inverted-U*	
	(1a) OLS	(1c) TWFE	(2b) Firm FE	(2c) + W
K^z	-0.92	-0.33^{**}		
	(0.73)	(0.13)		
W^z	2.84^{***}	2.00^{**}		
	(0.51)	(0.68)		
$K^z \times W^z$	1.95	-0.56^{*}		
	(2.36)	(0.31)		
Ecom			0.44^{*}	0.03
			(0.23)	(0.27)
Ecom^2			-0.028^{**}	-0.005
			(0.012)	(0.012)
Glassdoor				9.87^{***}
				(1.80)
Firm FE	No	Yes	Yes	Yes
Year FE	No	Yes	No	No
N	131	131	216	190
Firms	13	13	17	17

Dependent variable: ACSI score. Cluster-robust standard errors (at firm level) in parentheses; p-values use $t(G-1)$ degrees of freedom. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$. K^z and W^z are standardised CapEx/Revenue and Glassdoor overall, respectively. Ecom is national e-commerce share (%). Turning point for (2b): 7.9%. Attenuation of Ecom^2 in (2c) is consistent with the model's prediction that e-commerce affects satisfaction through workforce depletion.

workforce investment per se. A pre-period placebo (2015–2017, when e-commerce grew modestly) shows no effect ($p = 0.40$), confirming that the wage buffer activates only under substantial digital pressure. The effect operates through employment quantity, not wage levels: using wage growth as the dependent variable yields a null result ($p = 0.98$), consistent with the model's capacity-multiplier channel.

A caveat on the role of COVID-19: the e-commerce share jumped from 10.7% (2019) to 14.0% (2020), contributing substantial variation to the interaction term. This COVID-driven acceleration represents forced adoption rather than strategic digital investment, which differs from the model's framing of F as a managerial choice. The robustness of the result when COVID years are excluded ($p = 0.014$) mitigates this concern and confirms that the effect is not driven solely by pandemic-era variation.

Tables 5 and 6 present the full regression results.

6.4. Assessment

Three empirical strategies across two levels of analysis support the model's qualitative predictions. First, the inverted-U between e-commerce and ACSI ($p = 0.028$) and the post-shock ACSI decline in the event study ($p = 0.037$) both support the Front-Stage Trap. Second, the county-level wage–e-commerce interaction

Table 6 County-level regression results: Good Jobs Buffer.

	(1) Main	(2) Excl. COVID	(3) Urban	(4) Rural	(5) Placebo
$W^z_{c,2015} \times \text{Ecom}_t$	0.0026^{**}	0.0031^{**}	0.0050^{***}	0.0008	0.0023
	(0.0011)	(0.0013)	(0.0018)	(0.0014)	(0.0027)
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	30,951	24,768	15,532	15,419	9,321
Counties	3,115	3,115	1,557	1,558	3,115
R^2_{within}	0.008	0.013	0.042	−0.000	0.001

Dependent variable: $\ln E_{ct} - \ln E_{c,2015}$ (log employment growth relative to 2015 baseline). Standard errors clustered at county level in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$. Ecom_t main effect absorbed by year FE (national variable); $W^z_{c,2015}$ absorbed by county FE (time-invariant). Column (2): excludes 2020–2021. Columns (3)–(4): split at median baseline employment. Column (5): 2015–2017 only (pre-period placebo; 24 county-year observations dropped due to missing BLS data in 2015–2017). Negative within-R^2 in column (4) indicates the interaction term does not improve fit beyond fixed effects in the rural subsample, consistent with the buffer mechanism operating primarily in urban retail markets.

($p = 0.025$, $N = 30{,}951$, $R^2_{\text{within}} = 0.008$) provides a well-powered test of the Good Jobs Buffer at the employment-capacity margin; the low R^2 is expected for a single interaction term in a fixed-effects panel; comparable magnitudes are standard in large-sample panel studies with rich fixed effects (e.g., Chetty et al. 2009 report within-R^2 of 0.01–0.04 for tax-salience effects on consumer behaviour) and reflects a noisy reduced-form test of a structural model. County and year fixed effects absorb the vast majority of employment variation; the within-R^2 captures only the incremental contribution of a single interaction term net of 3,115 county intercepts and 10 year dummies. The between-effects complementarity sign ($+1.95$, $p = 0.41$) is directionally consistent with Proposition 4 but provides no statistical evidence given the 13-firm sample. The empirical contribution is one of directionally consistent evidence from the three significant tests, across different identification strategies, levels of analysis, and outcome variables, subject to the limitations in §7.2.

7. Discussion

7.1. Contributions and Managerial Implications

The three theorems together provide the formal micro-foundation that connects Ton's (2014) four operational choices to each other through the dynamics of K_{eff}. Her prescription to "operate with slack" corresponds to the positive workforce investment rate f: operational slack keeps W from depleting under moderate overload, and Theorem 2 quantifies the CLV return to doing so. "Standardise and empower" corresponds to a lower overload sensitivity σ: well-designed processes reduce the per-unit depletion rate and enlarge the safe upgrade zone of Theorem 3. "Offer less" maps onto keeping F below F_{crit}: the model gives this demand-side lever a computable threshold rather than a qualitative prescription. (Ton's prescription targets SKU variety and product-range complexity; in this model, F indexes front-stage digital feature complexity. The mechanisms are analogous—both increase back-stage demand—though not identical.) "Cross-train" maps onto a higher complementarity parameter p: cross-trained workers raise effective capability $h(W)$ at any given engagement level, increasing the marginal return to both technology and workforce investment simultaneously (Proposition 4). The model extends Oliva–Sterman with customer trust, a front-/back-stage domain split, and digital scaling: three extensions absent from the quality-erosion framework, now formalised in a single model that, to the best of our knowledge, provides the first dynamic formalisation of the service profit chain (Heskett et al. 1994) and Good Jobs Strategy (Ton 2014) with computable boundary conditions.

Managerial implications. Three implications follow. First, the maximum safe rollout speed is a workforce constraint (Theorem 3): the firm should verify that $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ exceeds the planned upgrade size. Second, workforce well-being is a leading indicator: W reaches crisis levels 6–21 months before trust does (EC Table EC.4). Third, the more digital the firm becomes, the more it should invest in its people (Proposition 4): conventional ROI calculations underestimate returns because the dominant capacity-multiplier channel is invisible in standard dashboards (Theorem 2).

7.2. Limitations and Directions for Future Research

Model limitations. The model treats F as exogenous; endogenising it would enable optimal deployment sequencing. The natural formulation is a Hamilton–Jacobi–Bellman (HJB) problem with F as control and (K, W, T) as states, but the piecewise-smooth switching at $G = 0$ makes the value function non-smooth across the switching manifold, requiring viscosity-solution techniques rather than standard smooth HJB methods (Filippov 1988). A tractable starting point would restrict attention to bang-bang controls ($F \in \{0, F_{\max}\}$) and characterise the optimal switching surface in (K, W, T)-space. The demand function $D(T, F) = (1 + aT)F^b$ grows without bound in F; it does not incorporate market saturation, which would moderate demand at very high digitalisation levels. The model's predictions are therefore most reliable near the Trap threshold, where super-linear demand growth is empirically grounded. The omission of a direct $W \to D$ channel (whereby disengaged workers reduce throughput directly, independent of the service gap) makes our results a conservative lower bound on true complementarity. The linear form $h(W) = (1 - p) +$

pW is a tractability choice; a concave specification (e.g., $h(W) = (1-p) + pW^{\gamma}$, $\gamma < 1$) would strengthen the Trap by making capability more sensitive to well-being at low W levels.

Empirical limitations. Three concerns warrant emphasis. First, with only $N_c = 13$ firm clusters, the firm-level analysis is at the boundary of asymptotic validity (Cameron et al. 2008); the county-level analysis ($N_c = 3{,}115$) does not share this limitation but tests a different margin (employment resilience rather than customer satisfaction). Second, Glassdoor ratings exhibit low within-firm temporal variation (within-firm SD is only 30% of total SD), making the complementarity interaction difficult to identify within firm over time. Store-level panel data with exogenous technology rollout events (e.g., staggered click-and-collect deployment across stores of the same chain) would permit sharper separation of the capacity-multiplier channel from confounds; the Harvard Shift Project worker survey offers a promising path with store-level technology exposure and well-being variables. Third, the county-level analysis uses retail wages as a workforce-investment proxy and employment as the outcome; it does not directly test the customer-satisfaction channel central to the model.

Extensions. Three, in order of expected impact: (i) store-level identification of p using technology rollout timing as instruments; (ii) a competitive multi-firm extension where workforce poaching creates strategic complementarity; (iii) stochastic demand forcing, which would transform the Burnout Cliff into a probabilistic threshold.

8. Conclusion

Standard operational dashboards measure the front-stage domain that invests in technology, not the back-stage workforce that fulfils the resulting demand and absorbs the consequences. This disconnect is not merely a measurement problem; it is a structural feature of omnichannel operations that makes cross-domain value destruction structurally predictable from two observable managerial choices: the workforce investment rate f and the digitalisation level F. This paper provides the formal instrument for both.

Three results follow that, to our knowledge, no existing single formulation produces. The Front-Stage Trap (Theorem 1): a computable digitalisation threshold $F_{\text{crit}} \approx 1.11$ beyond which additional front-stage investment destroys CLV by overloading back-stage workers, a sign reversal impossible in single-domain models.

The Good Jobs Buffer (Theorem 2): workforce investment raises CLV through a capacity-level channel confirmed exact by the $p{=}0$ counterfactual, which leaves CLV invariant to workforce spending and isolates the entire gain to service capacity. The Burnout Cliff (Theorem 3): the maximum safe upgrade size ranges from $\Delta F = 0.18$ for a workforce-neglecting firm to $\Delta F = 0.49$ for an investing firm, an approximately 172% difference; at baseline calibration, the neglected firm retains approximately 37% of the safe deployment headroom.

These results give operations teams two computable numbers standard dashboards do not report: a digitalisation ceiling (F_{crit}) and a maximum safe rollout size ($\Delta F_{\text{collapse}}$). Three empirical strategies provide directionally consistent evidence across different levels of analysis and identification strategies: an inverted-U between e-commerce and ACSI, a post-shock satisfaction decline following omnichannel launches, and county-level evidence that higher retail wages buffer employment against e-commerce growth ($N = 30{,}951$ county–year observations, 3,115 counties, $p = 0.025$). Store-level identification of p using exogenous technology rollout events remains the most important empirical follow-up. The broader implication extends beyond omnichannel retail: wherever automated front-stage processes generate demand on human back-stage operations, workforce investment is an operational prerequisite for digital strategy, not an optional cost line. The capacity-multiplier structure applies wherever technology adoption generates hidden back-stage load—healthcare, financial services, logistics—and the core mechanism is the same: front-stage automation that looks productive on its own ledger can destroy system-level value when the workforce constraint is ignored.

References

Acimovic J, Graves SC (2015) Making better fulfillment decisions on the fly in an online retail environment. *Manufacturing & Service Operations Management* 17(1):34–51.

Aral S, Brynjolfsson E, Wu L (2012) Three-way complementarities: Performance pay, human resource analytics, and information technology. *Management Science* 58(5):913–931.

Argote L, Epple D (1990) Learning curves in manufacturing. *Science* 247(4945):920–924.

Bainbridge L (1983) Ironies of automation. *Automatica* 19(6):775–779.

Bakker AB, Demerouti E (2007) The Job Demands-Resources model: State of the art. *Journal of Managerial Psychology* 22(3):309–328.

Bell DR, Gallino S, Moreno A (2018) Offline showrooms in omnichannel retail: Demand and operational benefits. *Management Science* 64(4):1629–1651.

Bijmolt THA, Broekhuis M, de Leeuw S, Hirche C, Rooderkerk RP, Sousa R, Zhu SX (2021) Challenges at the marketing-operations interface in omni-channel retail environments. *Journal of Business Research* 122:864–874.

Cameron AC, Gelbach JB, Miller DL (2008) Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics* 90(3):414–427.

Chetty R, Looney A, Kroft K (2009) Salience and taxation: Theory and evidence. *American Economic Review* 99(4):1145–1177.

Corbett CJ (2024) OM Forum: The operations of well-being: An operational take on happiness, equity, and sustainability. *Manufacturing & Service Operations Management* 26(2):409–430.

Demerouti E, Bakker AB, Nachreiner F, Schaufeli WB (2001) The Job Demands-Resources model of burnout. *Journal of Applied Psychology* 86(3):499–512.

Dubé A, Reich A, Shierholz H, West M (2020) Minimum wage shocks, employment flows and labor market frictions. *Journal of Labor Economics* 38(4):1009–1052.

Filippov AF (1988) *Differential Equations with Discontinuous Righthand Sides*. Dordrecht: Kluwer Academic Publishers.

Fisher ML, Gallino S, Netessine S (2021) Setting retail staffing levels: A methodology validated with implementation. *Manufacturing & Service Operations Management* 23(6):1562–1579.

Fornell C, Johnson MD, Anderson EW, Cha J, Bryant BE (1996) The American Customer Satisfaction Index: Nature, purpose, and findings. *Journal of Marketing* 60(4):7–18.

Gans JS, Goldfarb A (2026) O-ring automation. NBER Working Paper No. 34639. National Bureau of Economic Research, Cambridge, MA. Accessed March 2026.

Gupta S, Lehmann DR, Stuart JA (2004) Valuing customers. *Journal of Marketing Research* 41(1):7–18.

Heskett JL, Jones TO, Loveman GW, Sasser Jr WE, Schlesinger LA (1994) Putting the service-profit chain to work. *Harvard Business Review* 72(2):164–174.

Hübner A, Amorim P, Fransoo JC, Honhon D, Kuhn H, Martinez de Albeniz V, Robb D (2021) Digitalization and omnichannel retailing: Innovative OR approaches for retail operations. *European Journal of Operational Research* 294(3):817–819.

KC D, Terwiesch C (2009) Impact of workload on service time and patient safety: An econometric analysis of hospital operations. *Management Science* 55(9):1486–1498.

Kesavan S, Lambert SJ, Williams JC, Pendem PK (2022) Doing well by doing good: Improving retail store performance with responsible scheduling practices at the Gap, Inc. *Management Science* 68(11):7818–7836.

Kremer M (1993) The O-ring theory of economic development. *Quarterly Journal of Economics* 108(3):551–575.

Krusell P, Ohanian LE, Ríos-Rull JV, Violante GL (2000) Capital-skill complementarity and inequality: A macroeconomic analysis. *Econometrica* 68(5):1029–1053.

Martinez-Moyano IJ, McCaffrey DP, Oliva R (2014) Drift and adjustment in organizational rule compliance. *Organization Science* 25(1):44–65.

Mas A, Moretti E (2009) Peers at work. *American Economic Review* 99(1):112–145.

McKnight DH, Choudhury V, Kacmar C (2002) Developing and validating trust measures for e-commerce: An integrative typology. *Information Systems Research* 13(3):334–359.

Milgrom P, Roberts J (1990) The economics of modern manufacturing: Technology, strategy, and organization. *American Economic Review* 80(3):511–528.

Mittal V, Ross WT, Baldasare PM (1998) The asymmetric impact of negative and positive attribute-level performance on overall satisfaction and repurchase intentions. *Journal of Marketing* 62(1):33–47.

Oliva R, Sterman JD (2001) Cutting corners and working overtime: Quality erosion in the service industry. *Management Science* 47(7):894–914.

Oliva R, Sterman JD, Giese M (2003) Limits to growth in the new economy: Exploring the 'get big fast' strategy in e-commerce. *System Dynamics Review* 19(2):83–117.

Rahmandad H, Repenning N (2016) Capability erosion dynamics. *Strategic Management Journal* 37(4):649–672.

Repenning NP, Sterman JD (2001) Nobody ever gets credit for fixing problems that never happened. *California Management Review* 43(4):64–88.

Saghiri S, Wilding R, Mena C, Bourlakis M (2017) Toward a three-dimensional framework for omni-channel. *Journal of Business Research* 77:53–67.

Sirdeshmukh D, Singh J, Sabol B (2002) Consumer trust, value, and loyalty in relational exchanges. *Journal of Marketing* 66(1):15–37.

Sterman JD (2000) *Business Dynamics: Systems Thinking and Modeling for a Complex World*. Boston: Irwin/McGraw-Hill.

Ton Z (2014) *The Good Jobs Strategy: How the Smartest Companies Invest in Employees to Lower Costs and Boost Profits*. New York: Houghton Mifflin Harcourt.

Tucker AL (2007) An empirical study of system improvement by frontline employees in hospital units. *Manufacturing & Service Operations Management* 9(4):492–505.